

Roger Ni

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EDUCATION

University of California, San Diego

Sep. 2024 – Jun. 2028

B.S. in Mechanical Engineering,

La Jolla, CA

- Coursework: MATLAB, Statics, Dynamics, Intro to Materials, Solid Mechanics, CNC Machining

EXPERIENCE

IT Technician

Aug. 2025 – Present

UC San Diego

San Diego, CA

- Resolved network infrastructure anomalies, account access issues, university compliance and hardware faults for a campus community exceeding 70,000 users.
- Optimized user experience by collaborating with multidisciplinary team members to streamlining communication for non technical users, achieving an efficient 20 minute average resolution time

Undergraduate Research Assistant

Dec. 2025 – May 2026

Carbide Materials Research, Xtreme Materials Lab, UC San Diego

San Diego, CA

- Conducted research under Professor Olivia Graeve by analyzing peer-reviewed literature and extracting carbide material properties to support development of a 400+ entry materials database and reference website
- Compiled and validated approximately 50 material entries, organizing experimental data and material properties into a structured database for research use

EXTRACURRICULARS

Rocket Propulsion Laboratory

May. 2026 – Present

Project Phoenix – Hardware Avionics Engineer

San Diego, CA

- Migrated a legacy Arduino rocket flight system to a compact ESP 32 architecture, consolidating ground station and flight computer functionalities
- Engineered physical interfaces for high-speed Ethernet communication MOSFET-driven valve controls, and pressure transducer data acquisition while reducing system size by 30%

Sep. 2025 – May 2026

Project Daedalus – Rocket Design Team Member

San Diego, CA

- Co-engineered design, manufacture, and launch a solid-fuel G-class model rocket with a simulated apogee exceeding 4,000 ft
- Modeled a nose cone and fin-can assembly in SOLIDWORKS and fabricated flight components using PLA 3D printing
- Executed OpenRocket flight simulations to evaluate aerodynamic stability and mass distribution, achieving a 1.6-caliber stability margin and predicted maximum velocity of 650 ft/s
- Designed and assembled a custom Arduino-based avionics system integrating inertial and pressure sensors, an onboard camera, and microSD storage for flight data acquisition and recording

Engineering Without Borders – Project Tijuana

Jan. 2025 – Jun. 2025

Team Engineer

San Diego, CA

- Co-engineered with six members to fabricate and assemble three load-tested community benches and custom 3D-printed toys for children's daycares in Tijuana
- Operated woodworking equipment including multi-cutters, bandsaws, and sanders to fabricate structural components according to engineering drawings, dimensional tolerances, and specified component placement

PROJECTS

Remote-Controlled Soccer Robot

Sep. 2025 – Dec. 2025

Mechanical Design Project

San Diego, CA

- Co-engineered and fabricated a remote-controlled soccer robot, using Fusion 360 to develop the drivetrain, ball collection, and launching mechanisms under energy and material constraints
- Integrated motors, gears, belt drives, and fabricated acrylic and metal components, iterating mechanical prototypes through rigorous testing cycles, significantly improve mechanical reliability and system responsiveness

TECHNICAL SKILLS

Software: SolidWorks, Fusion 360, MATLAB, OpenRocket, C, Java Microsoft Excel

Manufacturing: LaserCAMM, , 3D Printing, Circuit Soldering, Woodworking, CNC Manufacturing, Lathe, Milling